

The Arf Switches Off: Quadratic Forms over the Full Nimber Field

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Abstract

Over every perfect characteristic-2 field with surjective Artin–Schreier map, every nonsingular finite-dimensional quadratic form is hyperbolic and the quadratic Witt group is zero. Conway’s full nimber field On_2 is algebraically closed of characteristic 2, so this applies to On_2 . A possibly singular form is classified by the dimensions of the space, its polar radical, and its quadratic radical; there is no further Arf coordinate. The Clifford/Brauer–Wall class of every nonsingular form is therefore split.

The finite-field Arf bit remains correct as a *relative* invariant over the chosen finite nim-subfield. It dies after a quadratic scalar extension, so its directed colimit through the finite nim-subfields is zero. This also exposes an empty mathematical middle case: every ordinal below ω^ω is algebraic over $\mathbb{F}_2$, so every finite metric in that segment has a common finite field of definition. The backend detects that field only when the required Kummer excess rows are verified and the checked degree arithmetic remains representable. Finally, Ogdoad’s partial **Ordinal** arithmetic need not pretend to construct Artin–Schreier roots that lie outside its verified window, even though the full- On_2 classification is mathematically complete.

1 Introduction and main result

Let F be a field of characteristic 2, let V be finite-dimensional, and let

$$Q : V \longrightarrow F, \quad Q(av) = a^2Q(v),$$

be a quadratic form. Its polar form is

$$B(u, v) = Q(u + v) + Q(u) + Q(v).$$

It is alternating. Write $R = \text{rad}(B)$, and call Q nonsingular when $R = 0$, as in the characteristic-2 terminology of Elman–Karpenko–Merkurjev [5, Ch. II, §7]. Its quadratic radical is

$$\text{rad}(Q) = \{r \in \text{rad}(B) : Q(r) = 0\}.$$

For the fields considered below this is the kernel of a linear functional on $\text{rad}(B)$. We call a characteristic-2 field *quadratically closed* in this note when it is perfect and its Artin–Schreier map is surjective. In repository terms this is an ordinary characteristic-2 (q, b) metric: the general-bilinear deformation coordinate **metric.a** is empty. The Artin–Schreier map and quotient are

$$\wp : F \longrightarrow F, \quad \wp(t) = t^2 + t, \quad F/\wp(F).$$

For a nonsingular form and a symplectic basis $(e_i, f_i)_{i=1}^r$, the usual characteristic-2 Arf class is represented by

$$\sum_{i=1}^r Q(e_i)Q(f_i) \pmod{\wp(F)}.$$

Over a finite field this quotient has two elements and the absolute trace identifies it with $\mathbb{F}_2$. Over the full number field no finite absolute trace exists, so the intrinsic Artin–Schreier quotient is the correct starting point.

The answer is: *nothing replaces it over the full number field*. The class already lives naturally in $F/\wp(F)$, and this quotient is zero when $F = \text{On}_2$. More strongly, we give the complete singular normal form below.

Both the field theory and the form theory used here are standard; the content of this note is their application to the full number field and the corrective observation that an Arf bit is relative to a declared subfield and dies in the colimit. Conway’s nim addition and nim multiplication turn the class of ordinals into the algebraically closed characteristic-2 field On_2 (*On Numbers and Games*, Theorem 49) [3]; Lenstra states this explicitly and analyzes the algebraic-closure beginning segment [7]. The surrounding quadratic-form theory is standard; see also Baeza [2] and the characteristic-2 treatment of Hoffmann–Laghribi [6]. Risnoveanu’s recent work on quadratically closed number fields is adjacent to, rather than competing with, the full-field application here [8].

2 Elementary splitting lemmas

Lemma 2.1 (Quadratically-closed consequences). *Let F be a quadratically closed characteristic-2 field in the sense above. Then Frobenius $x \mapsto x^2$ and the Artin–Schreier map $\wp(x) = x^2 + x$ are both surjective, and $F/\wp(F) = 0$.*

Proof. Perfection makes Frobenius surjective, and Artin–Schreier surjectivity is part of the hypothesis. The quotient conclusion is immediate. $\square$

Corollary 2.2. *Every algebraically closed field of characteristic 2 is quadratically closed in the sense above.*

Proof. For $a \in F$, algebraic closure supplies roots of $X^2 - a$ and of $X^2 + X - a$, respectively proving Frobenius and Artin–Schreier surjectivity. $\square$

Let H denote the hyperbolic plane: in a basis e, f ,

$$Q(e) = Q(f) = 0, \quad B(e, f) = 1.$$

Lemma 2.3 (Every nonsingular plane is hyperbolic). *Let F be a characteristic-2 field with surjective Artin–Schreier map. Every nonsingular quadratic plane over F is isometric to H .*

Proof. Choose a symplectic basis e, f , so $B(e, f) = 1$, and put $a = Q(e)$, $b = Q(f)$. Thus

$$Q(xe + yf) = ax^2 + xy + by^2.$$

If $a = 0$, then e is a nonzero isotropic vector. If $a \neq 0$, choose $t \in F$ satisfying

$$t^2 + t + ab = 0$$

and set $v = (t/a)e + f$. Direct substitution gives

$$Q(v) = \frac{t^2 + t + ab}{a} = 0.$$

In either case take a nonzero isotropic vector v . Nonsingularity supplies w_0 with $B(v, w_0) = 1$. For every $c \in F$,

$$Q(w_0 + cv) = Q(w_0) + c^2Q(v) + cB(w_0, v).$$

Only now substitute the already established equality $Q(v) = 0$ and take $c = Q(w_0)$. Then $w = w_0 + Q(w_0)v$ satisfies $B(v, w) = 1$ and

$$Q(w) = Q(w_0) + Q(w_0)B(w_0, v) = 0.$$

Thus v, w are a hyperbolic basis. □

Proposition 2.4 (The nonsingular part). *Every nonsingular quadratic space of dimension $2r$ over a characteristic-2 field with surjective Artin-Schreier map is isometric to $H^{\perp r}$. Consequently $W_q(F) = 0$.*

Proof. An alternating nondegenerate form has even dimension and a symplectic plane. Lemma 2.3 makes the quadratic form on that plane hyperbolic. Its polar-orthogonal complement is again nonsingular. Induction splits off r hyperbolic planes. Since every nonsingular form is hyperbolic, its quadratic Witt class is zero. □

The polar radical carries a small residue of information, but it is not an Arf class. Let Z denote the one-dimensional zero form and let A denote the one-dimensional form $x \mapsto x^2$; both have zero polar form.

Lemma 2.5 (The radical has only two forms). *Let R be an s -dimensional vector space over a perfect field of characteristic 2, and suppose the polar form of $Q|_R$ is zero. Then there is a unique linear functional $\ell : R \rightarrow F$ such that*

$$Q(r) = \ell(r)^2.$$

Up to isometry, $Q|_R$ is $Z^{\perp s}$ if $\ell = 0$, and $A \perp Z^{\perp(s-1)}$ if $\ell \neq 0$.

Proof. For a basis $r_1, \dots, r_s$, choose the unique $c_i \in F$ with $c_i^2 = Q(r_i)$ and define

$$\ell\left(\sum_i x_i r_i\right) = \sum_i c_i x_i.$$

Because the polar form vanishes,

$$Q\left(\sum_i x_i r_i\right) = \sum_i x_i^2 Q(r_i) = \left(\sum_i c_i x_i\right)^2.$$

Uniqueness follows from injectivity of Frobenius. A nonzero linear functional admits a basis in which it is the first coordinate; this gives the asserted normal forms. □

3 Complete normal form over quadratically closed fields

Theorem 3.1 (Quadratically-closed characteristic-2 classification). *Let Q be a finite-dimensional quadratic form over a quadratically closed characteristic-2 field F , let B be its polar form, and write*

$$\text{rank}(B) = 2r, \quad \dim \text{rad}(B) = s, \quad \epsilon = [\text{rad}(Q) \subsetneq \text{rad}(B)],$$

where brackets denote the indicator of the displayed condition. Then

$$(V, Q) \cong \begin{cases} H^{\perp r} \perp Z^{\perp s}, & \epsilon = 0, \\ H^{\perp r} \perp A \perp Z^{\perp(s-1)}, & \epsilon = 1. \end{cases}$$

Here the second case necessarily has $s \geq 1$; if $s \geq 2$, its radical summand $A \perp Z^{\perp(s-1)}$ is isotropic. Thus two such forms are isometric if and only if the triples

$$(\dim V, \dim \text{rad}(B), \dim \text{rad}(Q))$$

agree. In particular every nonsingular form has a unique isometry class at fixed dimension, its Arf class is zero, and no transfinite Arf bit exists.

Proof. Choose a vector-space complement W to $R = \text{rad}(B)$. The restriction $B|_W$ is nondegenerate: if $w \in W$ is orthogonal to W , it is also orthogonal to R , hence lies in R , and therefore $w = 0$. Since $B(W, R) = 0$, the quadratic space is the orthogonal sum

$$(V, Q) = (W, Q|_W) \perp (R, Q|_R).$$

Proposition 2.4 identifies the first summand with $H^{\perp r}$, and Lemma 2.5 gives the two stated radical normal forms. In the proof of Lemma 2.5, $\text{rad}(Q) = \ker \ell$: thus it has dimension s when $\ell = 0$, and dimension $s - 1$ when $\ell \neq 0$. Equivalently, $\epsilon = [\text{rad}(Q) \subsetneq \text{rad}(B)]$. These dimensions recover the two normal forms and are isometry invariants. $\square$

Corollary 3.2 (Full nimber field). *The theorem applies to the full nimber field On_2 . In particular, every nonsingular finite-dimensional quadratic form over On_2 is hyperbolic and $W_q(\text{On}_2) = 0$.*

Proof. Conway's field is algebraically closed of characteristic 2, so Corollary 2.2 applies. $\square$

Remark 3.3 (No hidden proper-class step). The theorem concerns a finite matrix of coefficients and performs finitely many field operations and root choices. The fact that On_2 is a proper class causes no extra algebraic step: every polynomial used in the proof has an ordinal root in Conway's field. Explicitly, start with the set-sized subfield generated by the finitely many coefficients. Replacement, applied to its set of polynomials and the finite set of ordinal roots of each, shows that the relative algebraic closure inside On_2 is set-sized; the argument can be carried out there.

The theorem says exactly what survives from the existing **ArfInvariants** report after scalar extension to full On_2 :

$$\text{arf} = 0, \quad \text{rank} = 2r, \quad \text{radical_dim} = s, \quad \text{radical_anisotropic} = \epsilon.$$

The last field means “ Q is nonzero on the polar radical”; it is not a remaining anisotropic nonsingular core.

4 Why the finite Arf bit is still correct

There is no contradiction between Theorem 3.1 and the existing finite-nim-field classifier. Isometry depends on the ground field.

Let $K_d = \mathbb{F}_{2^d} \subset \text{On}_2$. For a nonsingular form over K_d , its Arf class lies in

$$K_d/\wp(K_d) \cong \mathbb{F}_2,$$

where the isomorphism is the absolute trace. A form of nonzero Arf class is not hyperbolic over K_d , but it becomes hyperbolic after scalar extension to K_{2d} . The finite-field antecedent goes back to Dickson [4].

Proposition 4.1 (Every finite Arf bit dies one stage later). *The scalar-extension map*

$$W_q(K_d) \longrightarrow W_q(K_{2d})$$

is zero. Hence the directed colimit of the finite-field quadratic Witt groups inside On_2 is zero.

Proof. For $c \in K_d$, the polynomial $X^2 + X + c$ either already has a root in K_d or is an irreducible quadratic and therefore has a root in K_{2d} . Thus every c lies in $\wp(K_{2d})$, so the induced map $K_d/\wp(K_d) \rightarrow K_{2d}/\wp(K_{2d})$ is zero. Arf’s 1941 classification theorem [1] then makes the Witt-group map zero. Every element at every finite stage therefore maps to zero at a later stage, which is precisely the vanishing of the directed colimit. $\square$

For the smallest example, the plane $Q(e) = Q(f) = B(e, f) = 1$ is anisotropic over $\mathbb{F}_2$. If $t^2 + t + 1 = 0$ in $\mathbb{F}_4$, then $te + f$ is isotropic, so the same plane is hyperbolic over $\mathbb{F}_4$. The finite Arf bit records failure to split over the declared finite field; it was never invariant under arbitrary number scalar extension.

This settles the proposed “direct-limit finite-subfield invariant”: the direct limit is canonically zero. It also distinguishes two legitimate questions:

- Over the ideal full scalar world motivating `Ordinal`, namely On_2 , Theorem 3.1 is the complete classification.
- If one deliberately freezes the smallest subfield K generated by the coefficients and restricts all basis changes to $\text{GL}_n(K)$, the relative invariant belongs to $K/\wp(K)$. For transfinite coefficients this K need not be finite or algebraically closed. This Arf class is generally not by itself a complete higher-dimensional isometry invariant over such a field. That is a different, explicitly relative classification problem; it is not a scalar-level invariant of On_2 and has no canonical reduction to one bit.

Ogdoad already makes the first distinction for finite metrics: the raw `ArfInvariants` result is evaluated at a separately detected degree, and the higher `ClassifyWitt/ClassifyBrauerWall` surfaces store that degree alongside the Arf bit. The degree is part of those classes, not decoration.

5 The Clifford and Brauer–Wall boundary

The quadratic Witt conclusion immediately answers the Clifford half of the problem, with one scope qualification.

Corollary 5.1 (The nonsingular Clifford class splits). *For every nonsingular finite-dimensional quadratic form Q over On_2 , the graded Morita class of $\text{Cl}(Q)$ is zero on Ogdoad’s characteristic-2 Clifford/Brauer–Wall surface.*

Proof. By Theorem 3.1, $Q \cong H^{\perp r}$. Orthogonal sum maps to graded tensor product of Clifford algebras. For a hyperbolic basis e, f , the relations

$$e^2 = f^2 = 0, \quad ef + fe = 1$$

identify $\text{Cl}(H)$ with the graded matrix algebra on a $1|1$ -dimensional space, which is graded Morita-trivial. Therefore

$$[\text{Cl}(Q)] = r[\text{Cl}(H)] = 0.$$

□

This corollary concerns the subgroup of the graded Brauer–Wall theory carried by Clifford algebras of nonsingular quadratic forms—the exact surface used by the repository’s finite characteristic-2 classifier. It does not claim a classification of every possible graded central simple algebra in characteristic 2. For singular Q , a nonzero polar-radical vector becomes a non-scalar homogeneous graded-central Clifford element (the characteristic-2 graded sign is trivial); if its square is zero, it also generates a nilpotent graded ideal. Thus the full Clifford algebra is not graded central simple on the required surface, so there is no Brauer–Wall class of the whole singular algebra; any polar-nondegenerate complement nevertheless has the split class above. Such a complement need not define a canonical quotient quadratic form when $Q|_{\text{rad}(B)} \neq 0$. The radical normal-form flag ϵ is quadratic-form data, not a Brauer–Wall coordinate.

6 Executable boundary

The mathematical ground field and the current Rust execution domain are not the same object.

- The intended scalar world is full On_2 , but `Ordinal::nim_mul` is implemented only through a source-verified Kummer window. Generic `Scalar::mul` is deliberately panic-on-escape.
- Mathematically, every ordinal below $\omega^{\omega^{\omega}}$ is algebraic over $\mathbb{F}_2$ and has finite degree. A finite metric there therefore lies in the compositum $\mathbb{F}_{2^m}$, where m is the least common multiple of the entry degrees. The repository’s `ordinal_common_finite_subfield_degree` certifies this field when every required Kummer excess is in the verified table and the checked degree arithmetic does not overflow, and returns `None` otherwise. Thus there is no mathematical third regime of entries in this segment that are not jointly contained in a finite nim-subfield, although the partial implementation can honestly report an operationally unknown degree.
- `arf_ordinal_finite` correctly classifies a metric over its detected and thereby declared common finite subfield, and uses that field’s absolute trace. It does not classify after scalar extension to the whole algebraic-closure segment. Its result is relative to the separately detected degree; the higher class wrappers record that degree, as Proposition 4.1 requires.
- A metric with coefficients outside the algebraic-closure initial segment, or an arbitrary metric over ideal full On_2 , is split over On_2 by Theorem 3.1; but constructing the isometry may require an Artin–Schreier root or square root outside the represented multiplication window. Returning a bare executable “`Arf = 0`” without that distinction would confuse a mathematical base-extension theorem with a checked in-window witness.

Accordingly no new root-search oracle is required to close the research problem. A future implementation may expose a theorem-level `classification_over_full_on2` report, or a checked isometry only when all needed roots are representable. It should not invent a transfinite trace to $\mathbb{F}_2$.

7 Conclusion

The classification has four consequences:

1. **Domain.** The ideal full-field classification motivating the `Ordinal` scalar is over On_2 ; finite or coefficient-generated subfields are explicitly relative refinements.
2. **Invariant.** Over full On_2 , the Arf/Artin–Schreier quotient vanishes. The complete possibly singular invariant is $(\dim V, \dim \text{rad } B, \dim \text{rad } Q)$.
3. **Root route.** Algebraic closure proves that the required roots exist. The proof gives explicit basis changes once such roots are supplied; executable root search remains bounded by the backend rather than by mathematics.
4. **Brauer–Wall.** Every nonsingular quadratic Clifford class is split; singular radicals carry no class of the same kind.

The full-nimber theory therefore closes by collapse, not by a new transfinite bit. The finite bit remains useful precisely because it remembers a smaller declared ground field.

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